Ohio University: Department of Physics and Astronomy

PHYS 6021 Quantum Mechanics II: Problem set 1 *due 7 February, 2024*

Zettili: Excs. 7.23 (use the lowering operator method from class also discussed near the bottom of p. 413), 9.8, 9.11, 9.16

And….

The ψ particle is a bound state of charmed quark c and its antiparticle $\bar{c}$. If we ignore the Coulomb potential, the $c\bar{c}$ interaction can be described by a spherical potential of the form $V = br$, where b is a constant with units of energy/distance. Using the Rayleigh-Ritz method, estimate the ground state energy of the $c\bar{c}$ system in terms of $b, \hbar$ and the quark mass. If the mass of each of the quarks is 1.5GeV, and the observed mass of the ψ is $3.095GeV$, estimate the constant b in units of GeV/fm ($1\text{fm} = 10^{-13}$ cm, $1\text{GeV} = 10^{9}eV$). Use as trial function $exp(-\alpha r^{3/2}/2)$, with α an adjustable parameter.